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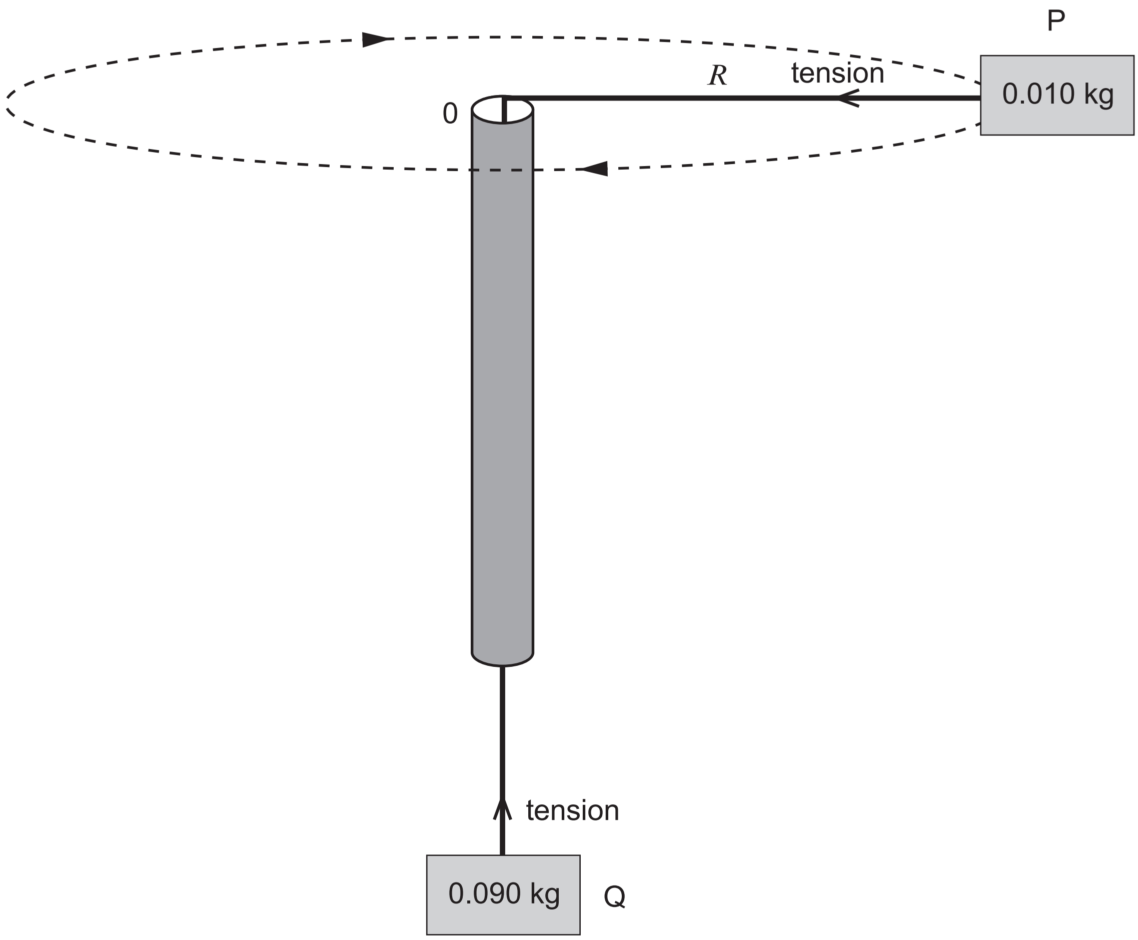
5. A piece of string is threaded through a hollow narrow cylinder. Two small objects, P and Q, with masses 0.010 kg and 0.090 kg respectively are attached to the ends of the string, as shown.

A student holds the cylinder and sets the 0.010 kg mass rotating in a horizontal circle of radius R, which is kept constant at 0.50 m. The time for 10 rotations is recorded. The tension in the string provides both the centripetal force on P and an upward force to hold Q in equilibrium.

The measurement is repeated for different values of R. All measurements are recorded in the table overleaf.



(a) Show that the speed, v, of mass P for each measurement is given by:

$v = \dfrac{2\pi R}{T}$ where T is the period of rotation. [1]

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1420U301
11

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(b) Complete the table. [4]

R / m	Time for 10 rotations / s	Period T / s	v / m s^{-1}	v^2 / m^2 s^{-2}
0.50	4.7			
0.60	5.2			
0.70	5.6			
0.80	6.0			
0.90	6.3			

(c) (i) Assuming that OP is horizontal, write an equation relating the centripetal force to v and R. [2]

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(ii) Hence, by using the equation for the forces acting on mass Q, show that:

$$v^2 = 9g\,R$$

where g is the acceleration due to gravity. [3]

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(d) (i) Use the data in the table to plot a graph of v^2 (y-axis) against R (x-axis). [4]

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(ii) Determine a value for g. [3]

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(iii) Suggest a way in which the experiment can be improved. [1]

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18

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SECTION A

Answer **all** *questions.*

1. *(a)* A car travels at a constant speed of $45.0\,\mathrm{km\,h^{-1}}$ around a curve in the road with a radius of $80\,\mathrm{m}$.

(i) Explain why the car is accelerating. [2]

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(ii) Calculate the angular velocity of the car (in $\mathrm{rad\,s^{-1}}$) as it travels around the curve. [2]

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(iii) Calculate the acceleration of the car and state its direction. [3]

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(b) Discuss how the application of science enables cars to travel safely around curves. [2]

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9

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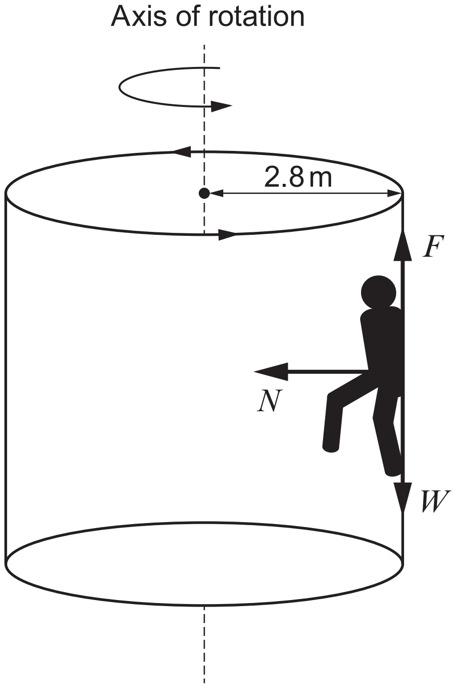
2. A fairground ride consists of a large empty cylinder of radius 2.80 m with its axis of rotation vertical. A person of mass 66.2 kg walks into the cylinder and stands against its inner wall. The cylinder is rotated around the axis. When the rotation rate is 36.0 revolutions per minute the floor is lowered, but the person remains fixed in place against the wall.

This is a student's diagram.



(a) Define the *radian*. [1]

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(b) Calculate:

(i) the period of rotation, T; [2]

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(ii) the speed, v, at which the person is moving in the circle. [2]

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(c) Three forces act on the person: the normal contact force from the wall, N, the weight, W, of the person, and the frictional force, F, from the wall. These are shown in the student's diagram.

Explain why the values of N and F are approximately 2 600 N and 650 N respectively. [4]

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(d) The maximum value that a frictional force can take is μN where μ is a dimensionless constant.

(i) Show that the person does not slide down the wall provided that μ is larger than approximately 0.25. [2]

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(ii) A student says that as the cylinder slows down the floor will need to be raised to support the person when the angular velocity reduces below a certain value. Justify this and determine this angular velocity if μ has a value of 0.45. [4]

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1420U301
07

15

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4. A satellite orbiting the Earth completes one revolution in 105 minutes.

Mass of the Earth = 6.0×10^{24} kg, radius of the Earth = 6400 km

(a) Explain what is meant by the term *radian*. [2]

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(b) Calculate the satellite's angular velocity, ω, in rad s^{-1}. [3]

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(c) The gravitational force on the satellite is given by $\frac{GM_E m}{R^2}$ where m is the mass of the satellite, M_E is the mass of the Earth and R is the radius of the orbit. Show that:

$$R = \sqrt[3]{\frac{GM_E}{\omega^2}}$$

[2]

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(d) Determine the height of the satellite above the surface of the Earth. [3]

(e) The mass of the Moon is 7.3×10^{22} kg and its radius is 1 740 km. Discuss whether a satellite would be able to orbit the Moon with a period of 105 minutes. [3]

1420U301
11

13

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2. (a) Explain what is meant by centripetal force and state its direction. [2]

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(b) A small sphere of mass 30 g at the end of a light string rotates in a horizontal circle of radius 0.80 m and completes 10 revolutions in 15 s.

(i) Show that the speed of the sphere is $3.35\,\mathrm{m\,s^{-1}}$. [2]

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(ii) Calculate the centripetal force on the sphere. [2]

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1420U301
03

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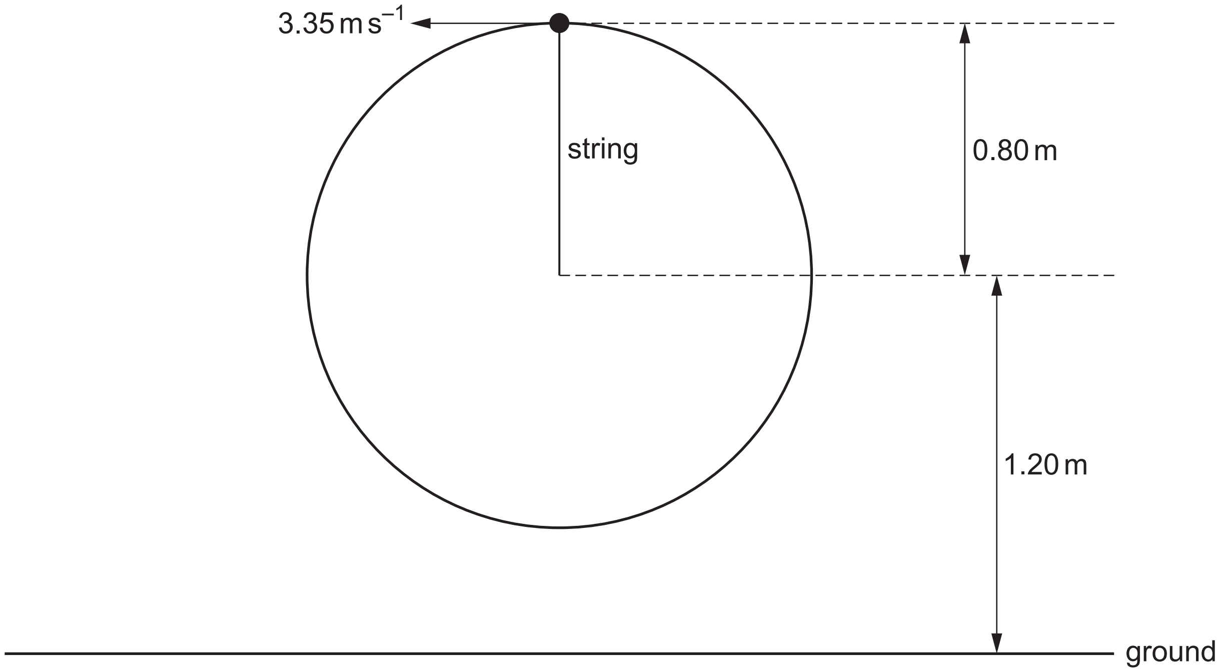
(c) The sphere now rotates in a **vertical** circle of the same radius, from a point that is 1.20 m above the ground. The speed at the top of the circle is $3.35\,m\,s^{-1}$.



Calculate the tension in the string at the **top of the circle**. [2]

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(d) (i) Calculate the speed of the sphere when it reaches the **bottom of the circle**. [3]

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(ii) A student claims that if the string breaks when the sphere is at the **top of the circle** it will reach the ground at a horizontal distance of approximately 2 m away from the point of release. Investigate if her claim is correct, justifying your answer. [4]

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1420U301 05

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2. (a) Explain what is meant by 'centripetal acceleration'. [2]

(b) The Earth has a radius of approximately 6370 km and rotates about its axis once each day. A student stands at the equator.

(i) Show that the angular velocity of the student is approximately 7×10^{-5} rad s^{-1}. [2]

(ii) Determine the centripetal acceleration of the student. [2]

(iii) I. Determine the speed of the student around the circular path. [2]

II. The student says, "this is a very high speed so my answer must be incorrect as I would feel the effects of this speed." Discuss whether or not the student is correct. [2]

10